



ZenCAM LITE Model Dash Camera Installation Guide



Preface

This *ZenCAM LITE Solution Installation Guide* is hereby prepared to better guide engineering personnel to install ZenCAM LITE and its accessories properly and quickly, and to improve installation efficiency.

This document mainly includes the following parts: preface, system overview, and preparation for installation, introduction to installation, and acceptance and cleaning.

This document is applicable to installation engineering personnel.

Important Notice

1. Before installation, please park the vehicle on the horizontal ground and shut down the engine (do not park the vehicle on a ramp or an inclined road).

2. Please read the section of packing list carefully and check carefully at the time of unpacking.

3. Please read the section of tool list carefully and provide installation tools before product installation.

4. Before installation, please observe the vehicle environment and follow the principles below:

a. The installation position and cabling of the product shall neither affect the driver's view nor affect the adjustment of the rearview mirror and sun visor.

b. The lens for monitoring the road condition ahead of the vehicle must be within the working range of the windshield wiper.

c. The installation position of the camera for monitoring the driver in the vehicle shall comply with local regulations.

d. The installation location should be convenient for Micro SD card and SIM card replacement and maintenance.

5. The appropriate installation position shall be selected according to the vehicle environment, and this document is for reference only.

6. The appropriate power supply connection method shall be selected according to the vehicle environment. **If loose wire connectors are adopted, connection to the power supply and all signal cables of vehicle is required, and shall be carried out by specialized personnel, as it may be dangerous for non-specialized personnel to operate the power system of the vehicle without authorization.** This document is for reference only.

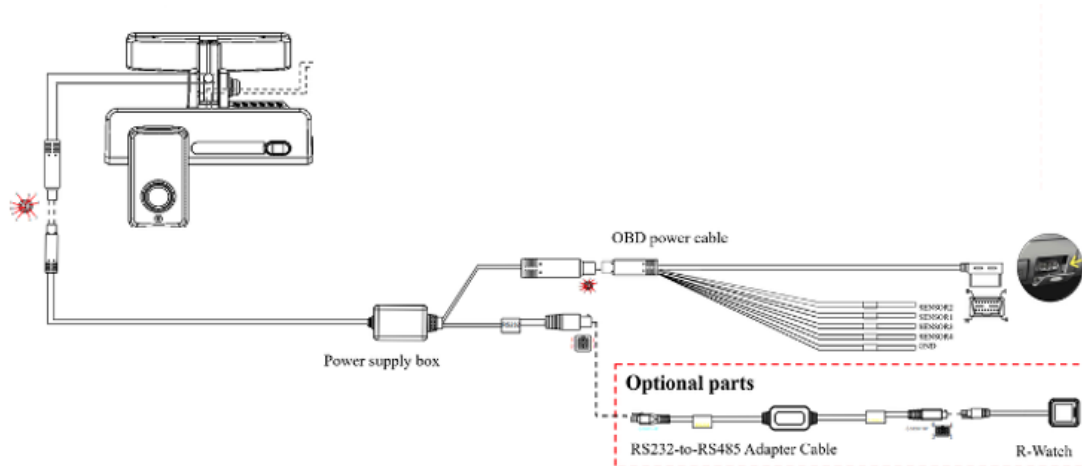
1. System Overview

1.1 Product Overview

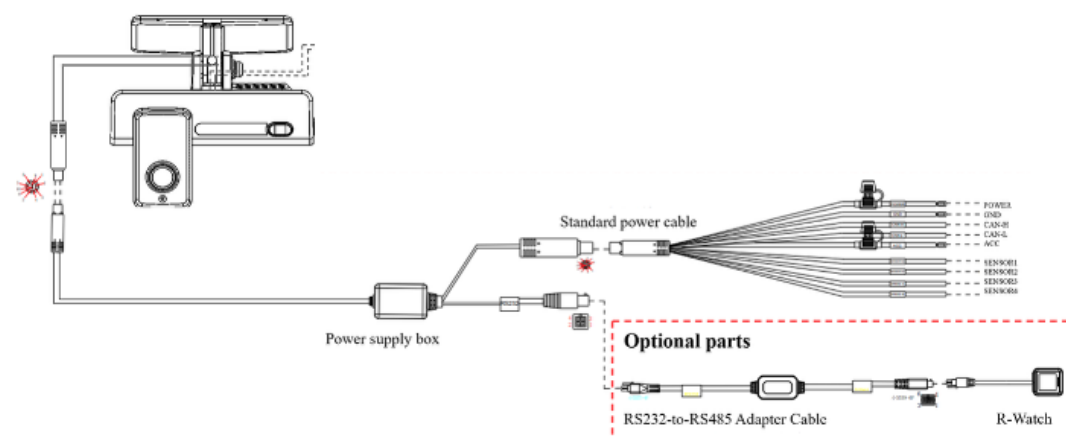
ZenCAM LITE is an AI dashcam that helps drivers to reduce traffic accidents and facilitates fleets to improve management efficiency. Based on AI technology, it can actively detect risky driving events and unsafe driving behaviors, supporting sending local real-time reminders to the driver to avoid risks and uploading events to the fleet management platform for driver training. It transmits real-time and accurate vehicle position information and operation data to the fleet management platform. It provides high-quality remote intercom and video live view playback to make the fleet management easier and more efficient.

The product is suitable for most weather conditions such as day and night, sunny and rain, and can be installed on buses, taxis, ordinary passenger cars, passenger vehicles, freight vehicles, dangerous goods transport vehicles, school buses, dump trucks, sanitation vehicles, and other vehicles.

1.2 Schematic Diagram of System Connection-Power Supply through OBD _ **Optional!**



1.3 Schematic Diagram of System Connection-Power Supply through Standard Wiring Harness / Power Cable (preferred method of installation)



2. Preparation for Installation

2.1 Technical Requirements for Installation

Relevant personnel shall be familiar with the functions, applications, and overall compositions of the product.

Relevant personnel shall understand the electrical circuits and structure of motor vehicles and common installation methods of in-vehicle devices.

2.2 Understanding of Installation Environment

Before device installation, relevant personnel shall have a clear understanding of the vehicle model concerned, the installation positions of the Dashcam and auxiliary cameras of the Dashcam, the type and length of cables required for each vehicle model, and the list of common auxiliary materials, so as to ensure successful completion of device installation and commissioning.

2.3 Confirmation of Vehicle Conditions and Vehicle-related Electrical Information

Confirmation of vehicle information is the basic precondition of successful installation and also the guarantee of division of responsibilities to avoid any damage to the vehicle. For each component, proceeding to next step is only allowed after clear confirmation, and each operation shall be confirmed by the person in charge of the vehicle and the installation personnel.

(1) Check the appearance and interior trims of the vehicle for any damage.

(2) Check whether the vehicle can ignite normally.

(3) Check whether the vehicle power supply system is in good condition.

*Note: Confirmation of the above information is crucial. Installation can only be carried out after the above information is considered normal through confirmation.

2.4 Power Supply Connection of Vehicle

ZenCAM LITE has two power supply connection modes:

1. Quick power supply connection through the OBD port: This mode is suitable for quick installation by users for self-installation but depends on OBDII power source.

2. **Power supply connection through the loose wire (preferred method):** This mode requires operation by specialized installation personnel. The following mainly describes the mode of connection to the vehicle power supply according to the power cable requirements of the product.

(1) Required tool: multimeter.

(2) Selection of power supply connection position

Note:

- When the vehicle is shut down, use a test pencil to detect whether the circuit is live. If it is live, it is judged as a constant power supply and then measure the voltage.
- When the vehicle is shut down and is in ACC position or ignition state, use a test pencil to detect whether the circuit is live. If it is electrically neutral in shutdown state, and is live in ACC position or ignition state, it is judged as an ACC power cable and then measure the voltage.

(3) Voltage measurement of power supply connection

- **Constant power supply:** When the vehicle is shut down, use a multimeter to measure whether the voltage of the constant power supply cable is about 24 V or 12 V. If the voltage of multiple cables is in the range, select the cable with higher current as the constant power supply connection cable.
- **ACC:** When the vehicle is in ACC position or ignition state, use a multimeter to measure whether the voltage is about 24 V or 12 V. If the voltage is 0 in shutdown state and 24 V or 12 V in ACC position or ignition state, select the cable as the ACC power supply connection cable.

*Note: During power supply connection, first conduct measurement at the positive and negative terminals of the power supply with a multimeter, to avoid wrong connection.

2.5 Connection of Necessary Signal Cables – CAN data and Turn Signal –

OPTIONAL / NOT STANDARD OR REQUIRED WHEN GPS AND AI / ADAS IS USED.

(1) CAN data cable - to obtain accurate vehicle speed (if GPS speed is used, the CAN cable may not be connected);

(2) Left and right steering signal cables - to obtain left and right steering information of vehicle; Please consult the maintenance engineer of the vehicle discipline for specific position of CAN data cable. Generally, the left and right steering signal cables are arranged on the fuse board below the steering wheel or below the front passenger dashboard, and measurement for these cables can be conducted using a multimeter.

*Note: If the measured signal is a pulse signal, the source of left steering/right steering/brake signal shall be set as pulse on the setting screen of the Dashcam; if the measured signal is a continuous high- or low-level signal, the source of left steering/right steering/brake signal shall be set as level on the setting screen of the Dashcam.

3. Preparation for Installation Material and Tool List

3.1 Packing List Inspection

After unpacking the product, please confirm whether the Dashcam is intact and whether the accessories are complete.

3.2 Preparation for Installation Tools

Before installation, the following installation accessories and tools shall be made available.

1



ZenCam LITE

2



Power supply box × 1

3



Standard power cable × 1
(with the fuse)

4



Allen key × 1

5



Mounting bracket × 1

6



Bracket bolt × 1

7



Crowbar × 1

8



Desiccant × 1

9



Alcohol cotton × 1

List of Installation Tools and Accessories				
S/N	Picture	Tool Name	Application	Quantity
1		Common screwdriver kit	Tighten screws, optional	1pcs
2		Crowbar	Pry up the vehicle panel	1pcs
3		Ties	Bundle cables	Prepare as needed
4		Dry cleaning cloth	Clean the dashboard	1pcs
5		Mobile phone/Pad	Install the Veyes App for video preview and parameter configuration	1pcs
6		Steel tape	Measure the installation height of the forward-facing ADAS lens and assist the installation in other scenarios	1pcs
7		Mark pen	Mark lines for Dashcam installation	1pcs
8		Cutting nippers	Cut and strip wires	1pcs
9		Insulated rubber tape	Wrap wire ends	1pcs
10		Scissors	Cut insulated rubber tape or wire clip	1pcs
11		USB flash disk	For future use	1pcs
12		Multimeter	Locate vehicle power supply	1pcs

			Measure the conduction of harness	
			Measure pulse signal	
13		3M adhesive tape	Fix DMS camera	1pcs
14		Three-legged ladder	Help to install the BSD camera	1
15		Waterproof sealant	Waterproofs backfill after punching	1
16		Waterproof tape	Waterproof protection for outdoor wire connectors	1

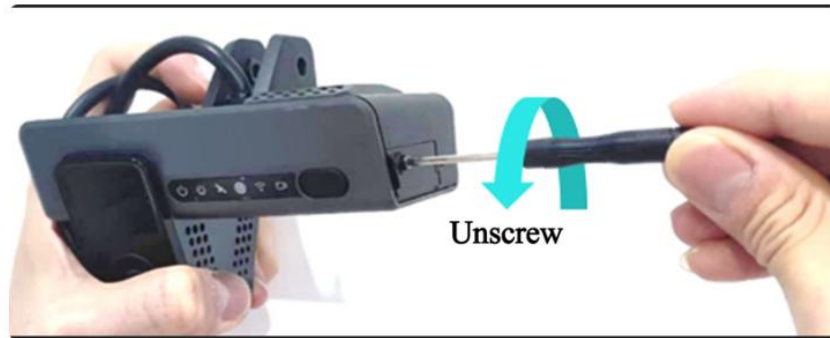
The following installation tools are also required for the installation of DMS camera if required:

DMS Camera Installation Tools				
S/N	Picture	Tool Name	Application	Quantity
1		Torsion drill	Tighten screws	1pcs
2		L-shaped Allen key	Adjust and fix the DMS lens at a certain angle (included in the DMS camera packaging)	1pcs
3		3.5 mm × 25 mm self-tapping screw	Fix camera; standard (included in the DMS camera packaging)	4pcs

4. ZenCAM LITE Installation

4.1 Installation of SIM Card and Memory Card ([setup and inserted prior to shipping](#))

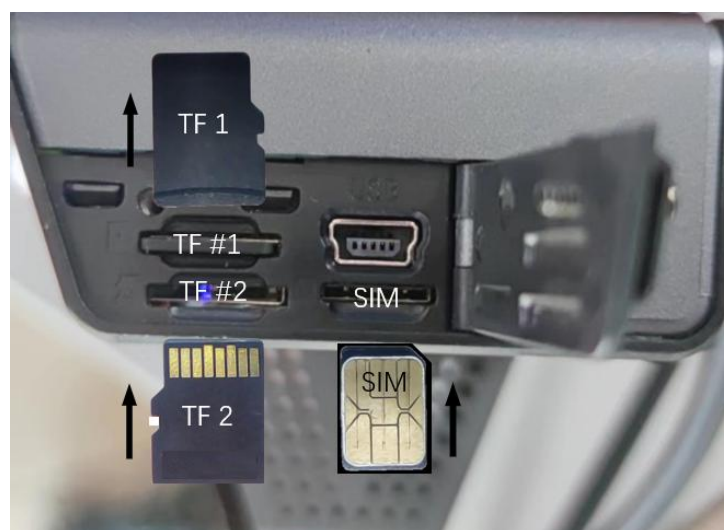
Take out the Dashcam (power-off) and use the Allen key in the kit to open the card slot panel on the right of the Dashcam by turning counterclockwise.



Install SIM card and Micro SD card as shown in the figure below (pay attention to the insertion direction of the cards). [This step is performed prior to shipping new equipment but may need to be done to reset SIM or replace the SD Card in cases where the camera is communicating a SIM or SD card error.](#)

If you feel smooth and flexible during installation and hear a clear sound of "Da" when pushing in the cards completely, it indicates that the cards are installed in the correct direction; if there is obvious friction resistance during installation, it indicates that the installation direction is wrong. Take out the cards in time to avoid any damage to the cards and the card holder.

If the Micro SD card and SIM card are too small to be pushed in the card slot completely by hand, you can put the card in the card slot and then push it in with the crowbar included in the package.



Note:

(1) Because the operating temperature range of the device is -40°C to +70°C, the Micro SD card and SIM card shall be able to function well in some harsh environments for a long time. Specifically, the metal contact on consumer-grade Micro SD cards and SIM cards is easy to oxidize under prolonged hot, humid, or salt-mist conditions. Frequent card pulling and pushing can lead to wear on the contact, and poor contact will also be caused by the deformed and bent ordinary cards due to the long-term pressure of the card holder. Therefore, if you use a SIM card, it is recommended to use an industrial plastic or ceramic M2M card for industrial devices (MP2/MP3/MS1 grade, operating temperature range -40°C to 105°C). If you use a Micro SD card, it is required to use an industrial-grade Micro SD card, which can be used in a wide operating temperature range (-40°C to 85°C) and has strong stability. HD quality -recommended Micro SD cards are preferred. Failure to use the appropriate card as required may damage the accessories or even the device.

(2) Do not touch the surface of the metal contact of the SIM card with hands when taking and installing the card, for fear of contaminating the SIM card by dust and sweat stain.

(3) Before installing the SIM card, please check the surface of the metal contact of the SIM card for any dirt (such as dust, fingerprints and water stains). If any, clean the surface with a piece of non-woven fabric or rubber.

(4) In TF card slot 1, the Micro SD card shall be pushed in with the metal strip side facing down; in TF card slot 2, the Micro SD card shall be pushed in with the metal strip side facing up.

- After installation of the SIM card and Micro SD card, fasten the card slot panel and tighten the screws.
- After installation of the SIM card and Micro SD card, remove the lens protection stickers on the front and rear lenses of the Dashcam, and tear off the protective film on the LED light on the front panel, as shown in the figure below.



4.2 Selection of Dashcam Installation Area

Requirements for installation area of ZenCAM LITE:

(1) **The Dashcam must be installed in the middle of the front windshield.** It is generally installed in the rearview mirror area above the centerline of the front windshield. A deviation less than 5 cm on the left and right sides is allowed for the installation position if it is not feasible to install the device in the middle as required (The deviation of the Dashcam relative to the centerline of the front windshield shall be calculated with the centerline of the forward-facing lens).

(2) When conditions permit, **the height of the DSC lens shall not exceed the height of the driver's eyes**, and the installation position shall be as low as possible provided that the driver's view is not obstructed. The linear distance from the position of the DSC lens to the driver's face shall not be more than 116 cm.

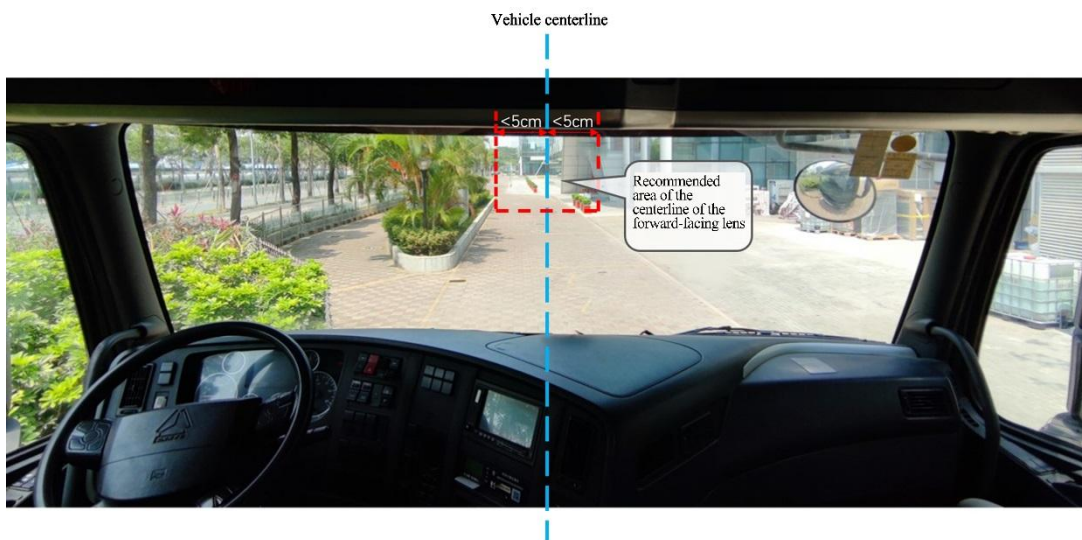
(3) The lens for road condition monitoring of the Dashcam must be within the working range of the left and right wipers (to ensure that the screen of the lens for road condition monitoring is clean and free from stains).

(4) The preferred vertical distance from the lens for road condition monitoring of the Dashcam to the ground is in the range of 4' – 8' or 130 cm-240 cm.

(5) Avoid installing other electronic devices around the Dashcam as far as possible, including ETC, intelligent rearview mirrors, and electronic tags; otherwise, they may affect the positioning signal of the device.

The installation position shall be determined in such a way that the Dashcam will not hinder the driver from viewing the front blind spot reflector, and there is no obstruction (such as interior rearview mirror or glass coating) within the field of view in front of and around the lens for cockpit monitoring and lens for road condition monitoring.

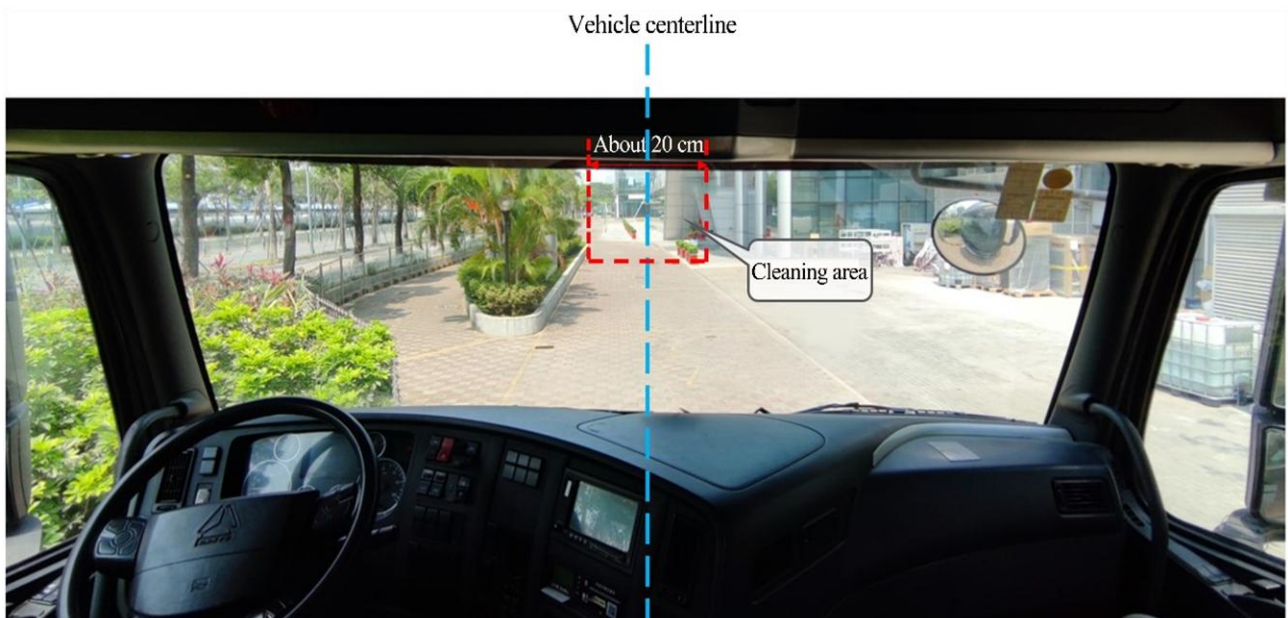
The installation area is generally selected as shown in the figure below:





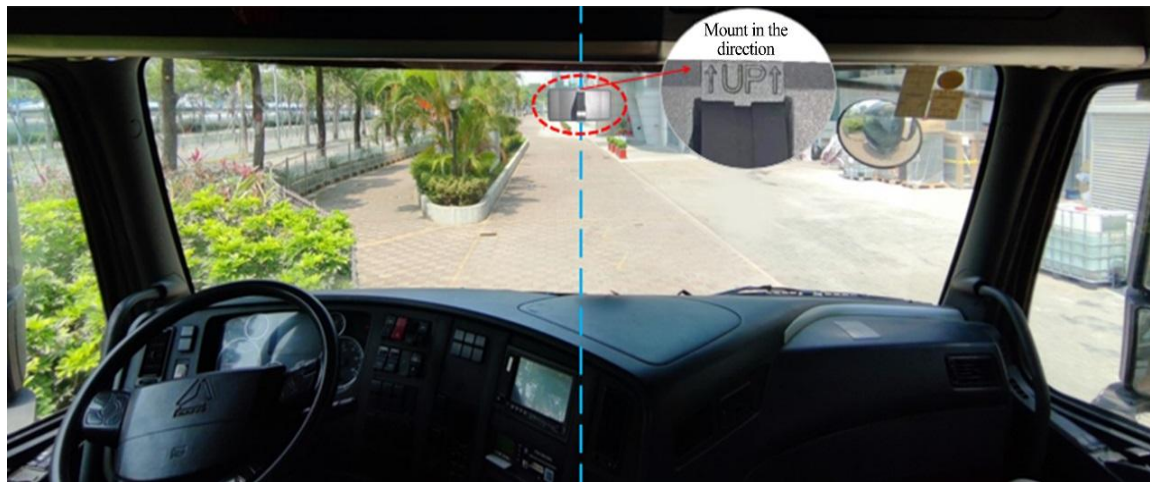
4.3 Installation of Dashcam Bracket

Park the vehicle on the horizontal ground, **clean the interior and exterior of the glass in the target installation area with alcohol cotton to ensure that no dirt on the glass in this area will affect the angle of view of the lens for road condition monitoring, and ensure the glass is dry.**



Tear off the 3M adhesive tape, install the bracket horizontally on the target installation area of the front windshield (the upper edge of the bracket should be parallel to the upper edge of the windshield) according to the direction indicated by the arrow on the bracket, and press the bracket for 10s to ensure that no air bubbles remain between the bracket and the glass.

Note: When installing the bracket, install the bracket according to the direction indicated on the bracket, so that the toothed side faces to the right.



4.4 Installation of Dashcam

Connect the Dashcam to the bracket with the front side facing inward (with the teeth on the right side of the bracket engaged with those on the right inner side of the Dashcam) and tighten the bracket stud clockwise with a PH2 cross screwdriver (before tightening, first adjust the Dashcam to be vertical).



4.5 Angle Adjustment and Fixation of Dashcam

Adjust the Dashcam back and forth so that it is vertical.

Fasten the bracket stud to ensure that the angle of the Dashcam will not be changed easily .

When fixing, ensure that the cockpit camera screen meets the following conditions:

- 1) The center of the cockpit shall be in the middle of the screen.
- 2) The cockpit screen shall be horizontal.
- 3) The vehicle steering wheel shall be shown at the lower left/right corner of the screen.

The cockpit rendering after the lens for cockpit monitoring is properly adjusted is as follows:



Use a screwdriver to fasten the bracket studs clockwise so that the Dashcam will not shake easily.

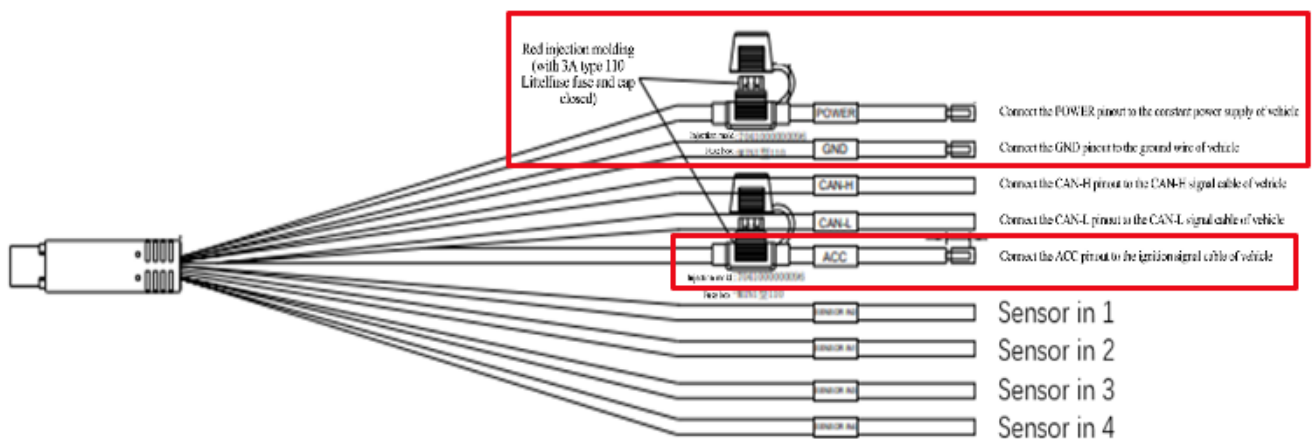
4.6 Power Supply Connection, Connection of Signal Cables and Cabling

4.6.1 Power Supply Connection

(1) If the mode of quick power supply connection through the OBD port is adopted, locate the OBD port of the vehicle and directly connect with the port.



(2) If the mode of power supply connection through the loose wire is adopted ([standard method](#)), according to the definition of power loose wire, connect **POWER/ACC/GND** with the power cable of the vehicle, respectively.



Note: The power line shall be connected using "special stripping-free connection terminal" where possible (no stripping is required, so as to avoid the risk of electric leakage), and the connection shall be wrapped with insulated rubber tape to avoid electric leakage/short circuit. If there is no special stripping-free connection terminal, stripped wires can also be used for connection. In this case, the connection process must conform to the standard specifications. After the connection is completed, the connection shall be wrapped with insulated rubber tape to avoid electric leakage/short circuit.

4.6.2 Connection of Signal Cables (CAN/Left/Right Steering Signal/Reversing)

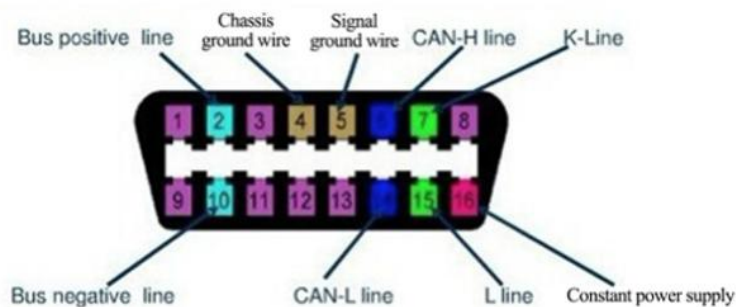
THIS STEP IS ONLY REQUIRED IF CUSTOMER CHOOSES TO USE THIS FEATURE – NOT STANDARD / OPTIONAL!

1. CAN

Consult the maintenance engineer of the vehicle discipline to locate the OBD port of the vehicle. Generally, the position of the OBD port of the vehicle is as shown in the figure below. Locate CAN-H and CAN-L cables of the vehicle behind the OBD port. Take the standard 16PIN inverted trapezoidal OBD port as an example, CAN-H and CAN-L cables generally correspond to pins 6 and 14, respectively. (The cable sequence varies with the shape of the OBD port. The example here is only for illustration.)

After the connection is completed, log in to the Veyes app to connect the ZenCAM LITE. Enter the configuration screen and set the CAN model and baud rate of the device and set the speed source as "OBD". At the same time, drive the vehicle for a short distance at the installation site to test the accuracy of vehicle speed pulse data. **This step can be done over-the-air by a Support Specialist @ Traxxis GPS Solutions, contact us @ 888.447.7059 ext. 101 for help and/or to confirm the installation.**

General Positions of OBD Port of Different Vehicle Models



2. Left steering/right steering/reversing signal – **NOT STANDARD, OPTIONAL!**

After locating the fuse board below the steering wheel or the front passenger dashboard, measure the cable corresponding to left steering/right steering/reversing signal according to the tips on the cover back of the fuse board or using a multimeter.

There are four IO signal cables in the standard loose wire, so left steering, right steering, and reversing signals can be connected. After connection, set the corresponding IO signal cable parameters through Veyes. If some customized loose wires have 8 IO signal cables, they should be connected as needed according to the actual application. **This can be done by a Support Specialist remotely.**

Note: If the measured signal is a pulse signal, the source of left steering/right steering shall be set as pulse on the setting screen of the Dashcam; if the measured signal is a continuous high- or low-level signal, the source of left steering/right steering shall be set as level on the setting screen of the Dashcam.

4.6.3 Cabling

Upon the completion of connection of main cables according to the schematic diagram of system connection, as well as power supply connection and connection of signal cables, arrange these cables using a crowbar according to the diagram below and conceal them in the interior trim panel or the panel of the dashboard (i.e. concealed cabling). If a DMS camera is required, the cable length of the DMS camera can be reserved for overall cabling.

(1) If the mode of power supply connection through the OBD port or loose wire is adopted, the cabling mode is as follows (make sure to hide and secure all cables in the vehicle prior to finishing):



Since ZenCAM LITE has a power supply box with a built-in turning-on/off control strategy, it is necessary to fix the power supply box at a certain position on the vehicle (**HIDE IT!**). Attention should be paid to the following items when the fixing position is selected:

- ① It should be close to the vehicle OBD port or loose wire power connection point / outlet.
- ② The mounting position should be flat.
- ③ It does not interfere with other components of the vehicle.
- ④ It keeps away from loudspeakers, engines, and other positions with excessive shake or vibration.
- ⑤ It should be as secluded as possible.

Due to the different positions of the OBD port of different vehicles, the corresponding cabling methods and the fixing position of the power supply box are also different.

Here is the recommended installation position for the power supply box. **You can also fix it in other positions according to the actual vehicle but be sure to hide it in the dash, out of reach.**

- Remove the side baffle of the driving seat, tear off the 3M tape on the power supply box, and fix it to the left or right baffle, as shown below:



Important Final Step:

Upon completion of installation process, call Traxxis GPS Solutions @ 888.447.7059 Ext. 101 to confirm installation and function. If any settings need to be changed or programmed, this can be done over the air through with the help of a Traxxis GPS Technical Support Specialist.